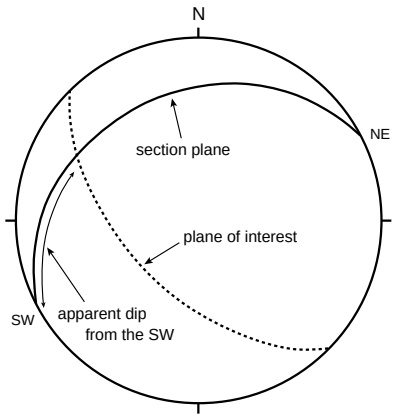
CONSTRUCTING CROSS-SECTIONS

Once you have chosen a cross-section plane orientation, putting the data on to the section involves two steps:
(a) the apparent dip of each planar reading (bedding, cleavages) on the cross-section needs to be determined;
(b) the projected position of each planar reading on the section needs to be determined.

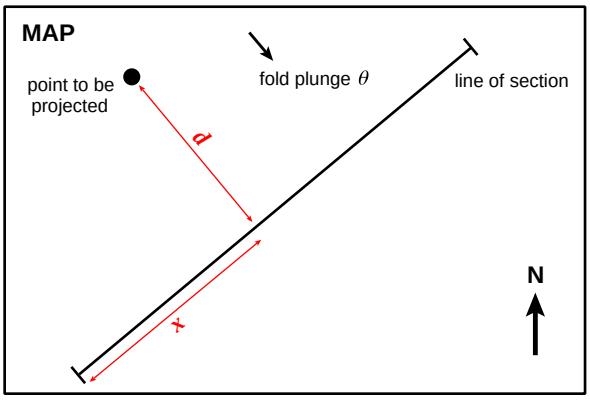
Evaluating apparent dips

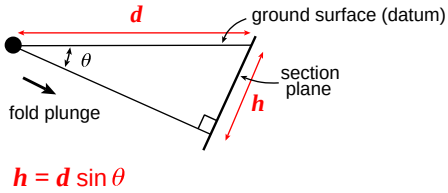
The easiest way to determine apparent dips is by using a stereonet. Note that for planes with a strike close to perpendicular to the line of section the difference between true dip and apparent dip is small



Projecting (flat topography case)

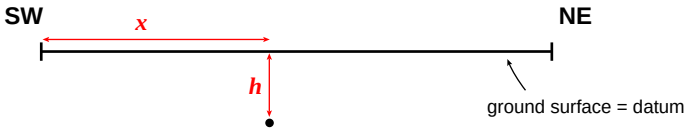
In an area of folded rocks, measurements that lie off the line of section should be projected parallel to the fold axis on to the line of section. In areas of flat topography this may be done using the simple trigonometrical procedure illustrated below.





$h = d \sin\theta$

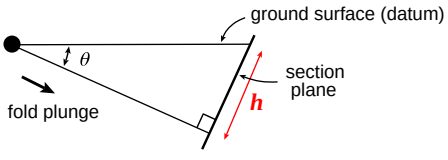
CROSS SECTION



Note that if we have a contact on the map, we can do this correction at several points along the contact. The resulting shape of the contact on the cross section has the correct apparent dip in the section plane

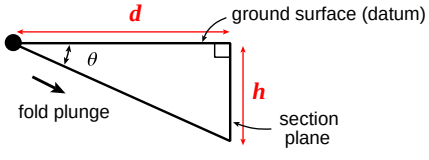
Note that the same process may be used to project data parallel to the fold axis on to a vertical cross-section:

Profile plane section



$h = d \sin\theta$

Vertical section

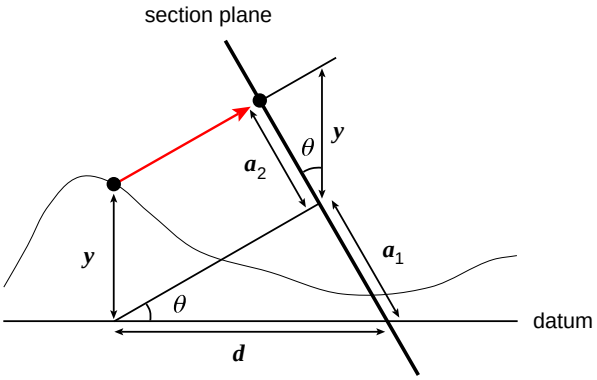


$h = d \tan\theta$

Projecting (rugged topography case)

When the topography is not flat the trigonometrical correction is more complicated because the elevation of the point of interest above datum must be accommodated:

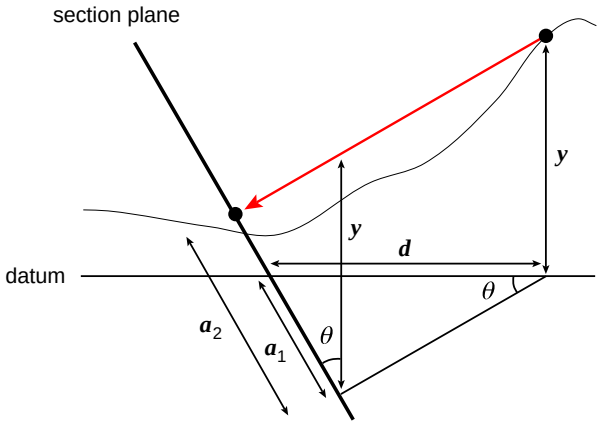
Readings that project up onto the section



For a reading at elevation y above datum

$h = a_1 + a_2 = d \sin\theta + y \cos\theta$

Readings that project down onto the section



For a reading at elevation y above datum

$h = a_2 - a_1 = y \cos\theta - d \sin\theta$

There is a method (method of arcs) that can be used to solve this trigonometry graphically.

Some general points when drawing cross-sections

The orientation of the cross-section

• In an area of uniformly tilted rocks, the ideal orientation for a cross section is a vertical plane oriented perpendicular to the strike of bedding. In this case all dips on the section are true dips.

• In an area of folded rocks, the ideal orientation for a cross -ection (assuming that it is designed to illustrate the geometry of the folds) is the fold profile plane, that is, such that one is looking along the fold axis. The section may be drawn looking up-plunge or down-plunge, whichever is the more convenient.

• Although the profile plane is the ideal orientation for a cross-section through folded rocks, in areas where the folds are only gently plunging, it is common practice simply to draw a vertical section. In this case the ideal orientation of the section is given by the strike of the profile plane.

The data to be shown on the cross-section

• Ideally, the cross section should be treated just like a field map: all the data used in its construction should be recorded on the section, that is, the exposure should be shown projected on to the plane of section, and the planar orientation measurements, presented as apparent dips in the section plane, should also be indicated.

• The cross section should be completed with the interpretation, that is, the positions of the stratigraphic boundaries, fold axial traces, and any faults should be shown using the same solid/dashed line conventions as on the map.

• In areas of simple structure it is usually convenient to show the topography but where the structure is more complex and much of the information has been projected on to the line of section from well away from it, this is unnecessary.

Presentation

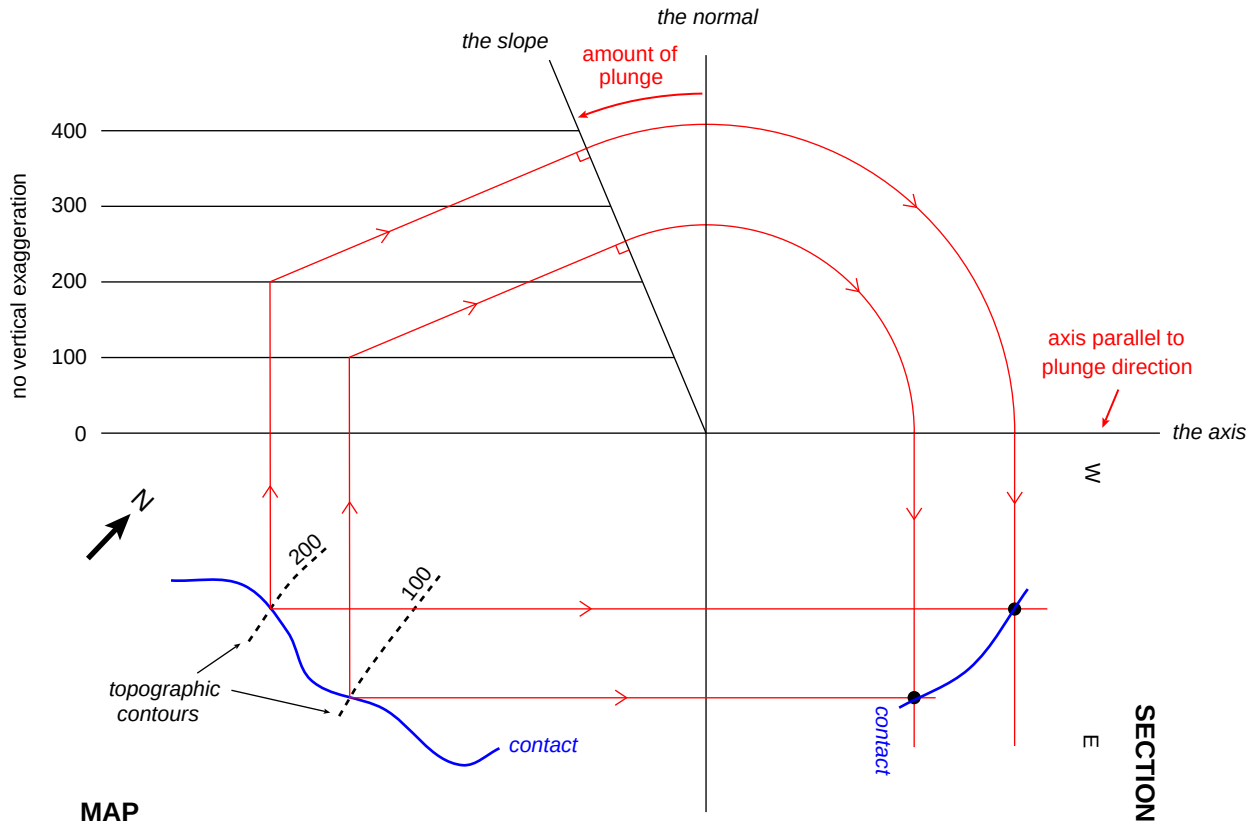
• Don’t forget a title with the strike/dip of the section plane included

• Don't forget to put orientation indicators on each end of the section. Include a scale bar for both the horizontal and vertical scales

• Include a comprehensive key to the symbols and, ideally, include a simple stratigraphic column

PROFILE PLANE SECTIONS USING THE ARC METHOD

The method of constructing a profile plane cross section using arcs is simply a graphical method for solving the set of trigonometrical calculations required when the map comes from an area of rugged topography.



The steps involved are

(1) Plot a stereogram to determine the plunge and plunge direction of the fold axis

(2) Draw a line (“the axis”) above the map parallel to the plunge direction. Draw another line (“the normal”) at right angles to this. Draw another line (“the slope”) at an angle to “the normal” equal to the plunge of the fold that passes through where “the axis” and “the normal” intersect

(3) Draw on the topography lines as shown above − make sure there is no vertical exaggeration

(4) You are ready to construct the section

You may find it convenient after Step 3 to draw a set of grid lines. Draw a set of verticals through the Section quadrant at a spacing of say, 5 mm on your piece of paper. Arc these around and down across the map. Now, instead of projecting points where the topographic contours cross you boundary, you can project the points where your grid lines cross the contact, although you will need to estimate the topographic height of these points to know how far up the vertical scale to go before projecting around.